Problem Chosen	2023	Team Control Number
A	**MCM/ICM**	**7391856**
	Summary Sheet	

Diversity as Insurance, Competition as a Filter

Summary

A species count alone cannot guarantee drought resilience. Diversity spreads the risk of planting unsuitable species, but competition can turn a diverse planting into a nearly single-species stand. We distinguish functioning abundance from conservation of all members.

We construct an annual seed-bank model with dormant survival, germination, fecundity and climate-dependent competition. Endpoints use an archived experiment on ten Mediterranean annual plants. A Markov process creates irregular drought with independently adjustable frequency and persistence. Future weather and management thresholds are scenarios, not observations.

All 1,023 nonempty species combinations receive equal initial seed totals and common weather. At drought probability .30, mean germinant density rises from 2,626 in random monocultures to 7,109 in random four-species plantings, then has a weak finite-sample peak near eight species at 8,579 germinants/m². Ten-species plantings nevertheless retain only about one species above the viable-seed threshold. For an illustrative 200-germinant floor, success fractions for one through four species are 40.0%, 66.7%, 83.3% and 92.9%. Four meets a 90% planning goal; five meets 95%; a selected monoculture also succeeds. "Four" is conditional composition insurance, not a biological constant.

Among the tested sets, the highest sampled fifth-percentile density shifts from *Sinapis alba* to *Medicago polymorpha* as drought probability rises from .30 to .60. At .30, clustering lowers *Sinapis alba*'s downside performance even when its mean rises; identical drought counts give different late-stage outcomes when order changes. Exact rare-growth conditions identify failure that extra competitors cannot repair. Uniform fecundity loss reduces this viable region. Homogeneous habitat loss reduces total abundance proportionally; demographic discreteness reveals extinction under joint pollution and area loss. A strict annual floor also requires more species than a long-window average.

Independent rational checks reproduce equilibria and invasion conditions, and bounds guarantee nonnegative, bounded trajectories. However, archived parameters do not reproduce the article's coexistence counts, and printed fecundities change high-richness predictions. We therefore recommend locally verified complementary responders, habitat and seed-bank refuges, pollution control, and separate monitoring of live density and species retention.

Keywords: seed bank; diversity insurance; competition; irregular drought.

Contents

1 From species count to an operational question 3

2 Assumptions and Justifications 4

3 Mechanism, units and measured inputs 4

3.1 Annual accounting 4

3.2 The measured parameter set and a source conflict 5

3.3 Irregular weather and matched comparisons 5

4 Analytical baseline and independent checks 6

4.1 What an insurance portfolio predicts 6

4.2 Exact equilibria, invasion and bounds 6

5 How much diversity helps, and what it fails to preserve 7

6 Species types and changing weather 9

7 Pollution, habitat and a boundary to diversity insurance 12

7.1 A specified pollution mechanism 12

7.2 Area loss and finite populations 12

8 Management for functioning and conserved communities 13

9 Strengths, limits and conclusions 14

Reproduction protocol 15

Report on Use of AI 17

1 From species count to an operational question

Grassland drought can arrive during an ordinarily wet season. The problem asks how community abundance and composition change through irregular weather, whether diversity helps, which types matter, how additional stresses alter survival, and what management should do. We define long-term viability in two distinct senses: a functioning stand produces living plants over repeated generations, while community conservation also retains its constituent species. Neither the number initially planted nor the size of a dormant seed bank establishes both outcomes.

Our focal system is an annual-plant community. A viable seed may germinate, remain dormant and survive, or die; a germinated plant produces the next cohort under crowding. This is appropriate for recurrent annual dynamics and a tractable empirical example. Perennial grasslands require an additional adult-survival stage. We use "adaptation" in the operational sense of population sorting and buffering under changing climate; genetic evolutionary adaptation is not identified by these data.

The question provides no measurements. We use the openly archived ten-species experiment of Matías et al. [1, 2]. Species were exposed to ordinary autumn rain or a 64-day delay of the first rains, corresponding to a 206 mm reduction in that experimental year. The archive provides germination, dormant-seed survival, low-crowding fecundity and focal-row interaction coefficients. These are measured endpoints for one location and season, not worldwide grassland constants.

We compared three routes. An equal-weight insurance portfolio is a useful baseline, but cannot explain displacement by competitors. A detailed soil-water/trait model could represent rooting depth and facilitation, but rainfall-to-soil-water, root overlap and recruitment functions are not identifiable from this dataset. We choose an annual seed-bank competition model because it links identifiable parameters to the requested mechanisms. We improve a fixed-climate equilibrium analysis by allowing irregular climate switching, testing chronological order, and adding demographic discreteness for finite habitat.

Table 1 gives direct answer locations. Model construction, code, numerical interpretation and writing used an AI assistant, disclosed in the appended report [8].

Table 1. Problem requirements and the corresponding test.

Requirement	Operational answer
Minimum and growth of richness	All subsets, common inputs; Sections 5 and 6
Species types	Vital-rate responses and rare-growth/invasion; Sections 4.2 and 6
Frequency and variation	Markov frequency/persistence and matched-count ordering; Section 6
Pollution and habitat	Fecundity-loss boundary and finite-area populations; Section 7
Viability and environment	Selection, refuges and monitoring with explicit limits; Section 8

2 Assumptions and Justifications

Shared premises and their modeling reasons are listed below. Weather comparisons and finite-population tests state additional local conditions.

1. **Annual seed-bank dynamics.** One reproductive cycle per year matches the annual experimental system. Perennial communities need an additional adult-survival stage.

2. **A closed, fixed pool.** Subsets of the ten parameterized species without immigration isolate initial composition. Dispersal and a universal species-count requirement remain outside this comparison.

3. **Pairwise competition.** Nonnegative focal-row coefficients identify pairwise competitive effects. Unmeasured facilitation, pollinator effects and higher-order interactions are excluded; we cannot attribute diversity benefits to those mechanisms.

4. **Homogeneous neighborhoods and common weather.** Representative areas and shared climate allow matched comparisons. They exclude local microclimates and real fragmentation geometry, which need separate measurements.

5. **Rates reused across generations.** Repeated endpoint rates permit annual scenarios. Time invariance and unchanged genotypes are model conditions; source uncertainty is tested, and new sites require re-estimation.

6. **Two-state weather.** A stationary Markov scenario separates drought frequency and persistence. The 200-year sequences are not fitted forecasts; frequency, clustering and chronological order are compared below.

3 Mechanism, units and measured inputs

3.1 Annual accounting

Let $N_{i,t}$ be viable seeds of species i before germination in year t, within a representative neighborhood of area $a_0 = \pi(0.075)^2 = 0.01767$ m². Let $g_i(z)$ be germination probability, $s_i(z)$ survival probability of a dormant seed, $\lambda_i(z)$ viable seeds produced per germinant without neighbors, and $\alpha_{ij}(z)$ the competitive effect of one germinant of species j on focal species i. With $G_{i,t} = g_i(z_t)N_{i,t}$, our recurrence is

$$N_{i,t+1} = \underbrace{(1 - g_i(z_t))s_i(z_t)N_{i,t}}_{\text{dormant survivors}} + \underbrace{\frac{\lambda_i(z_t)G_{i,t}}{1 + \sum_{j=1}^{S} \alpha_{ij}(z_t)G_{j,t}}}_{\text{new viable seeds}}. \tag{1}$$

This follows the annual-plant competition structure in the experiment [1]. Conspecific crowding appears once in the sum. Treating α as a coefficient per seed rather than per germinant would confound dormancy with competition.

The denominator is dimensionless. Germination and survival are probabilities, λ is seeds/germinant, and α has units germinant-1 in the fitted neighborhood. We report

$D_t = \sum_i G_{i,t}/a_0$ as germinant density and $B_t = \sum_i N_{i,t}/a_0$ as seed-bank density. The archive text has no unit annotation; the 7.5 cm neighbor-count radius is reported in the primary experimental methods [1, 3]. This area conversion assumes spatially homogeneous representative neighborhoods. It is not a validated biomass, cover or carbon conversion.

3.2 The measured parameter set and a source conflict

Table 2. Archive endpoints; W denotes normal rain and D delayed rain. Values are abbreviated for display; calculations use exact file decimals.

Code	λ_W	λ_D	g_W	g_D	s_W	s_D
BOOF	133.17	9.73	.0162	.0182	.091	.154
BRMA	20.25	40.57	.0568	.0819	.167	.151
CAAR	174.10	9.95	.0861	.0205	.171	.173
CABU	118.37	58.32	.0398	.0161	.289	.304
DIER	836.80	71.30	.0059	.0012	.324	.291
MACA	355.81	83.41	.0045	.0098	.319	.468
MEPO	1317.88	252.99	.0868	.0563	.213	.171
PARO	668.60	95.14	.0125	.0445	.031	.173
SIAL	2272.76	231.70	.0731	.0282	.269	.253
VISA	41.01	34.20	.1862	.2752	.216	.218

BOOF is *Borago officinalis*, BRMA *Bromus madritensis*, CAAR *Calendula arvensis*, CABU *Capsella bursa-pastoris*, DIER *Diplotaxis erucoides*, MACA *Matricaria chamomilla*, MEPO *Medicago polymorpha*, PARO *Papaver rhoeas*, SIAL *Sinapis alba*, and VISA *Vicia sativa*. The full 10×10 matrices are retained with the numerical source and results. The mirror file's MD5 equals the Dryad catalogue value, binding it to the archived file [2]; this does not establish agreement with every published table.

Several drought fecundities differ materially between the archived file and the article's rounded Table 1: PARO is 95.14 in the file but 501 in print; SIAL is 231.70 but 772. Furthermore our independent endpoint analysis finds two stable pairs in normal rain and none in delayed rain, whereas the article reports six and two. Transposing the matrix does not resolve the conflict and contradicts the archive's row-as-target instruction. We retain the file without silently substituting values. Section 6 tests the printed fecundities. Predictions from this parameterization must remain conditional until the discrepancy is resolved.

3.3 Irregular weather and matched comparisons

$Z_t \in \{0, 1\}$ denotes a normal or delayed-rain year. We use a two-state Markov chain with stationary drought probability p and persistence parameter ϕ:

$$\Pr(Z_{t+1} = 1 \mid Z_t = 0) = p(1 - \phi), \qquad \Pr(Z_{t+1} = 1 \mid Z_t = 1) = p + (1 - p)\phi. \tag{2}$$

Starting $Z_1 \sim \text{Bernoulli}(p)$ preserves the marginal probability p. With $\phi = 0$ years are independent; $\phi = 0.6$ creates runs of similar weather. Expected drought-run length is $1/[(1-p)(1-\phi)]$:

at $p = .3$, it increases from 1.43 to 3.57 years. The delayed-rain state explicitly represents drought during the normally rainy establishment season.

We test $p = .1, .3, .6$ and clustered versions of .3, .6, each with 48 synthetic 200-year realizations. The final 50 years form the assessment window. All 1,023 nonempty subsets receive 1,000 total viable seeds per neighborhood, equally divided among their initial species, and share each weather realization. A fixed-severity baseline sets $z = .3$ each year, interpolating g, s linearly and λ, α geometrically between endpoints. This is a structural interpolation, not an observed mild-drought treatment.

We enumerate subsets uniformly within each richness. Consequently the reported probability describes random composition and synthetic weather together; it is not the probability for an optimally selected mixture. We call a species quasi-persistent when its seed-bank density remains at least one viable seed/m² at the observation time. This threshold is a detection convention, not literal extinction in a continuous recurrence. A 200-germinant/m² planning floor is illustrative and can be replaced by a site's operational requirement.

4 Analytical baseline and independent checks

4.1 What an insurance portfolio predicts

For equally weighted species with common mean μ, variance σ^2 and pairwise response correlation ρ, the variance of mean performance is

$$\mathrm{Var}(\overline{X}_S) = \frac{\sigma^2}{S}\{1 + (S-1)\rho\}. \tag{3}$$

Thus adding species reduces weather variance when responses are imperfectly correlated; it cannot remove a shared catastrophe when $\rho = 1$. This baseline predicts insurance but no extra mean productivity and no competitive replacement. It also shows why functional response diversity is more relevant than merely counting nearly identical species. Field synthesis supports diversity-related resistance to climate extremes, but does not provide a universal recovery effect or calibrate our ten-species example [5].

4.2 Exact equilibria, invasion and bounds

Set $\beta_i = (1 - g_i)s_i$ and $\eta_i = \lambda_i g_i/(1 - \beta_i)$ for a fixed climate. A positive monoculture equilibrium requires $\eta_i > 1$ and has

$$G_i^* = \frac{\eta_i - 1}{\alpha_{ii}}, \qquad N_i^* = G_i^*/g_i. \tag{4}$$

For a candidate coexistence set, its positive equilibrium solves $AG^* = \eta - \mathbf{1}$. A species i invading a resident monoculture j has multiplier

$$m_{i|j} = \beta_i + \frac{\lambda_i g_i}{1 + \alpha_{ij} G_j^*}. \tag{5}$$

We require mutual invasion for the evaluated stable pairs and verify the positive fixed point and Jacobian spectral radius below one. For the archive parameters the normal-rain pairs are BOOF–

CABU and BOOF–PARO; no delayed-rain pair meets the same tests. This is an independent algebraic result for the file, not a replication claim for the article.

Positivity follows from nonnegative coefficients and denominators at least one. Since own-species crowding is included,

$$0 \le N_{i,t+1} \le \beta_i N_{i,t} + \lambda_i/\alpha_{ii}. \tag{6}$$

Here every measured $\beta_i < 1$, giving a finite scalar upper bound; no numerical population cap is imposed. An independent validator re-parses decimal inputs as exact rational numbers, checks scalar fixed-point substitution with exactly zero residual, solves pair equilibria, and matches invasion signs. It also checks finite outputs, planted-richness bounds, common initial inputs, matching weather counts and exact Markov stationarity. These are mathematical and implementation checks, not field validation.

At vanishing abundance, an isolated species has multiplier $r_i(z) = \beta_i(z) + \lambda_i(z)g_i(z)$. Its stationary long-run log growth is

$$\gamma_i(p) = (1 - p)\log r_i(0) + p \log r_i(1). \tag{7}$$

A necessary condition for persistence under purely competitive neighbors is $\gamma_i > 0$. The stationary marginal p, rather than ϕ, determines this isolated exponent; clustering still changes crowding histories and finite-population risk. For BOOF, $\gamma(.3) = .231$ but $\gamma(.6) = -.346$, so frequent drought prevents its isolated persistence. Adding competitors cannot increase its low-density growth in this model.

5 How much diversity helps, and what it fails to preserve

Figure 1 answers the richness question under a defined comparison. At $p = .3$, mean long-window germinant density increases 2.71-fold from random monocultures to random four-species plantings. In these 48 weather realizations, the gain from seven to eight species is only 0.87%; nine and ten reduce the mean slightly. The eight-versus-nine gap is only 41.9 germinants/m²; no stable statistical optimum is established. Hence the model predicts diminishing returns and a weak high-richness downturn, not unlimited proportional growth. Initial richness helps random-composition insurance; density gains flatten. Each point summarizes all subsets of that richness and 48 common-weather realizations. Right: an illustrative density floor and a 90% planning goal. The left panel of Figure 1 shows mean density; the right shows the fraction above the 200-germinant floor.

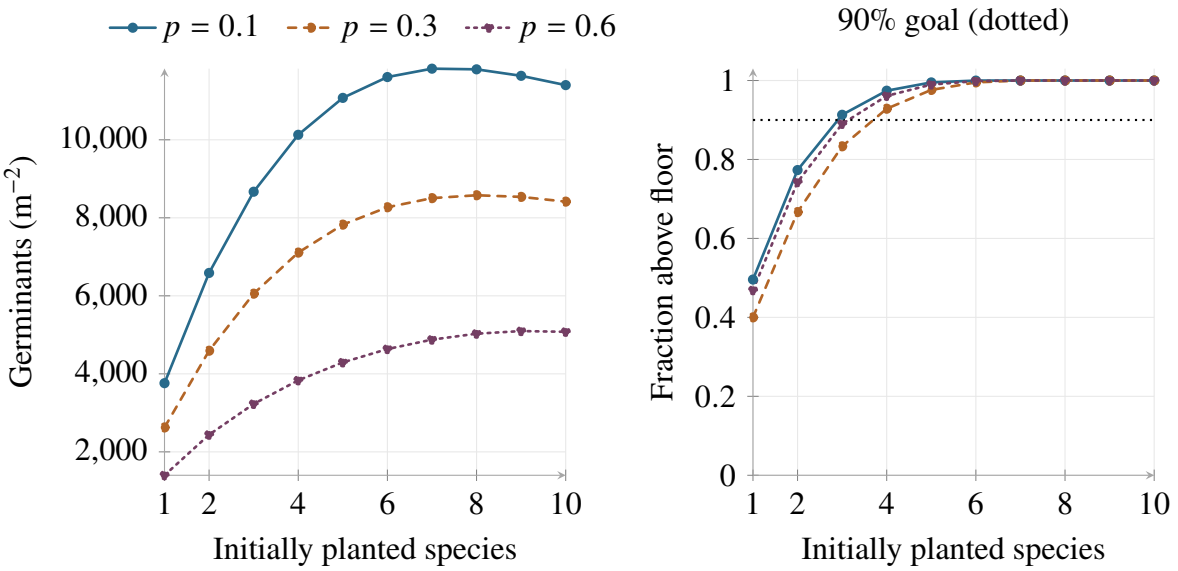


Figure 1. Richness and drought insurance

There is no unique minimum without a success criterion. One additional species already improves the average relative to random monoculture. For the 200-germinant floor at $p = .3$, success fractions are .400, .667, .833, .929, .976, .995, 1 for $S = 1, \ldots, 7$. Four is the smallest richness meeting 90%, five meeting 95%, and six meeting 99%. Every tested set with at least seven species clears the floor in these realizations. A selected SIAL monoculture also clears it, so seven is not necessary when composition is known.

These fractions closely follow selection insurance. Four of the ten species are individually productive enough under the moderate scenario; random planting draws at least one of these with probability

$$P_S = 1 - \frac{\binom{6}{S}}{\binom{10}{S}}, \qquad \binom{6}{S} = 0 \ (S > 6). \tag{8}$$

This gives $P_4 = 13/14$, exactly the observed moderate-scenario floor fraction. An independent combinatorial enumeration checks all ten richness values against the numerical fractions; this agreement identifies selection insurance rather than an unmeasured facilitation effect. It supplies a transparent explanation for the conditional "four" result. Changing the floor, species pool or competitive environment changes the answer. The minimum for a 90% long-window goal is one species at a floor of 50, four at 200 or 1,000, and six at 5,000 germinants/m². Requiring the floor in *every* final-window year instead of the 50-year average increases the minimum to five at 200 and seven at 1,000; no richness meets the annual 5,000 floor in these tests. Thus even the unchanged species pool has no single threshold independent of the management objective.

Function and biodiversity diverge. Left: species above the viable-seed threshold approach one even in initially rich plantings. Right: isolated log growth; blue is positive and red negative. Positive growth is necessary, not sufficient under competition.

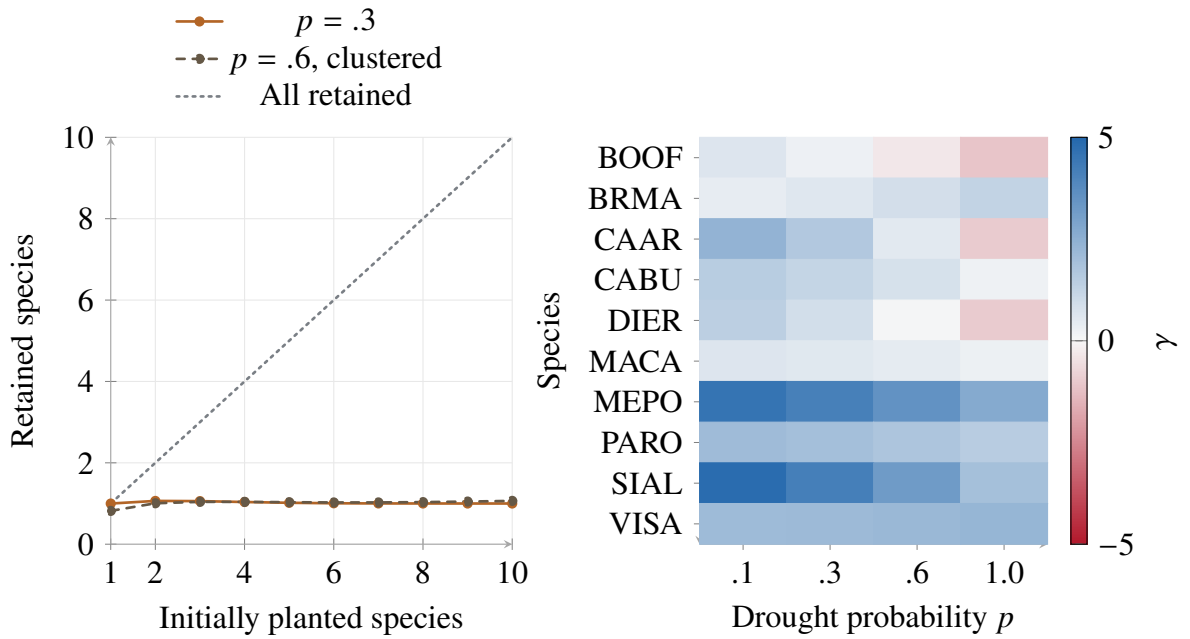


Figure 2. Species retention and isolated growth

Figure 2 exposes a stronger limitation. Ten-species plantings retain only approximately one species above the threshold in the moderate climate. Even the frequent clustered climate averages about 1.06 species, not ten. Initial diversity improves the chance of including a successful competitor, but it does not establish long-term preservation of diversity. The continuous model's very small positive populations are not counted as ecologically conserved species.

6 Species types and changing weather

The response types follow mechanisms, not family labels alone. BOOF and CAAR lose large fecundity and rare-growth capacity under delayed rain; BRMA raises both fecundity and germination, providing a different response. VISA combines higher germination in dry years with moderate fecundity and positive isolated growth in both climates. MEPO and SIAL produce many seeds but differ in competition sensitivity. The highest sampled fifth-percentile long-window density among all tested sets and these 48 weather realizations is SIAL alone at $p = .1$ and .3, and MEPO alone at .6. These are finite-sample candidate rankings, not statistically certified optima. Selecting the densest monoculture would nevertheless fail a biodiversity-conservation objective.

Trait measurements from the same experiment link climatic responses with phenotypic plasticity [3]. We use that evidence to motivate complementarity and local trait measurements, not to substitute a trait label for an invasion test. Seed-bank buffering can promote coexistence when species respond differently and competition covaries appropriately with their favored environments [4]; a dormant bank alone does not guarantee that storage effect. The strong displacement in our parameterization shows why this mechanism must be tested rather than assumed.

More frequent drought changes performance and composition, but its effect on the relative

insurance gain is not monotone. Under $p = .1$, random three-species sets clear the illustrative floor in about 91.3% of tests; at $p = .3$ they clear it in 83.3%; at $p = .6$ they clear it in 89.0%. The species favored by dry years change the productive pool. Low drought frequency therefore does not erase the importance of composition.

At $p = .3$, clustering lowers the selected SIAL monoculture's sampled fifth-percentile density from 7,958 to 6,915 germinants/m², while its mean rises from 9,630 to 10,535. Long wet runs increase mean production; long dry runs increase downside exposure. A mean-only ranking misses this tradeoff. Figure 3 illustrates one realization for initially ten-species communities; it is a scenario trajectory, not a forecast of a particular future century. Irregular years generate distinct population histories. The two weather processes have the same nominal drought probability .3, but different persistence. The middle panel shows species-specific seed-bank density for the clustered realization; values below one seed/m² are clipped to zero on the log scale. The trajectories share the model and initial total, not an identical weather sequence. The top panel of Figure 3 shows live density and the bottom panel weather states.

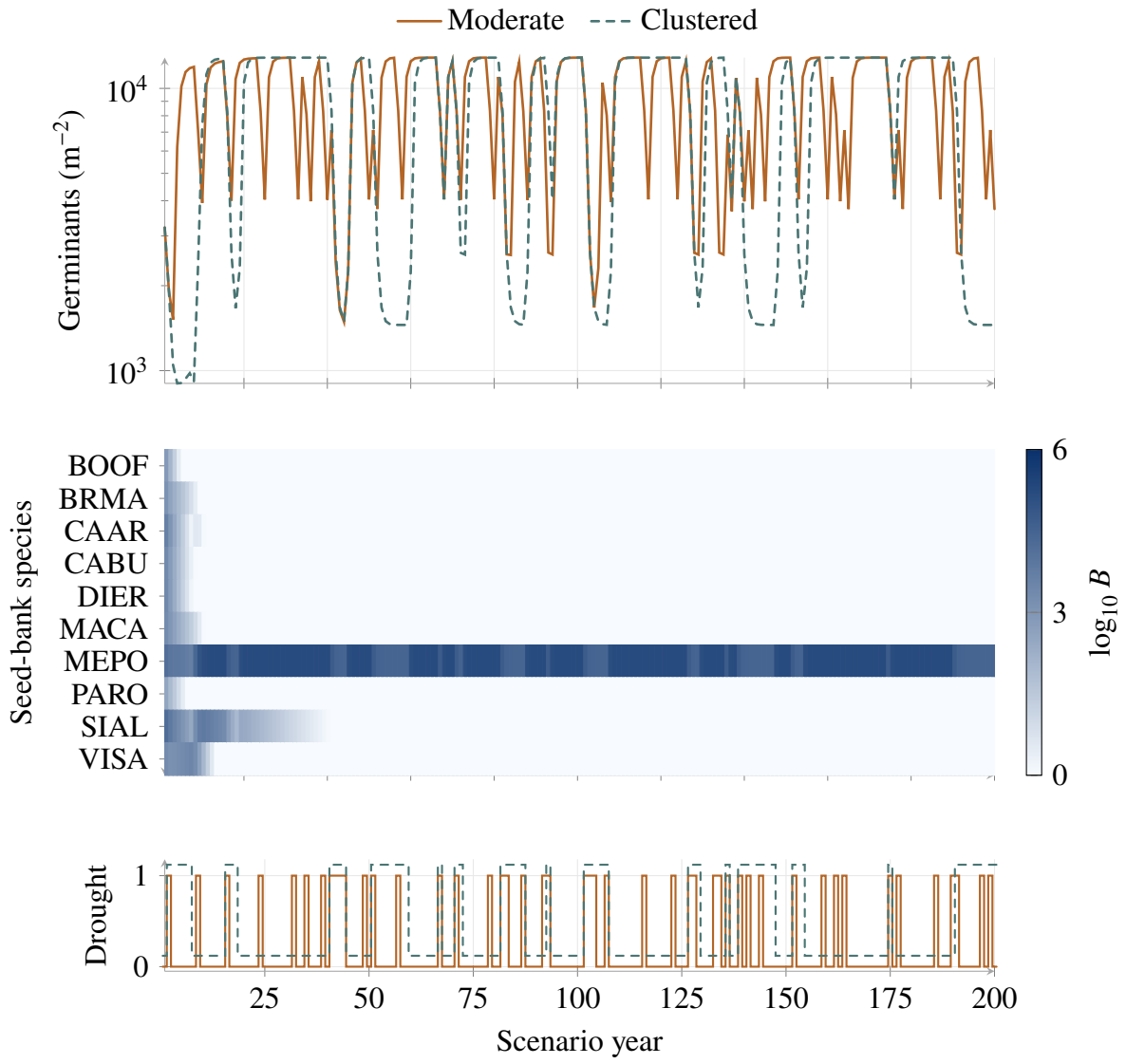


Figure 3. Population dynamics under irregular drought

We then remove a sampling confound: three deterministic sequences contain exactly 66 dry

and 134 normal years. The mean final-window density across all subsets is 6,742 for dispersed drought, 11,732 when drought occurs first, and 1,110 when it occurs last. The difference reflects order and the evaluation window, not a changed drought count. Neither a fixed average climate nor a cumulative drought count is a sufficient state description. Figure 4 compares these source and ordering tests. The left panel uses exactly the same 24 weather realizations for the archive and printed fecundities; the constant-severity curve is a separate structural comparison. The right panel shows different late-stage densities for equal drought counts. This concerns finite-window function, not an invariant asymptotic claim.

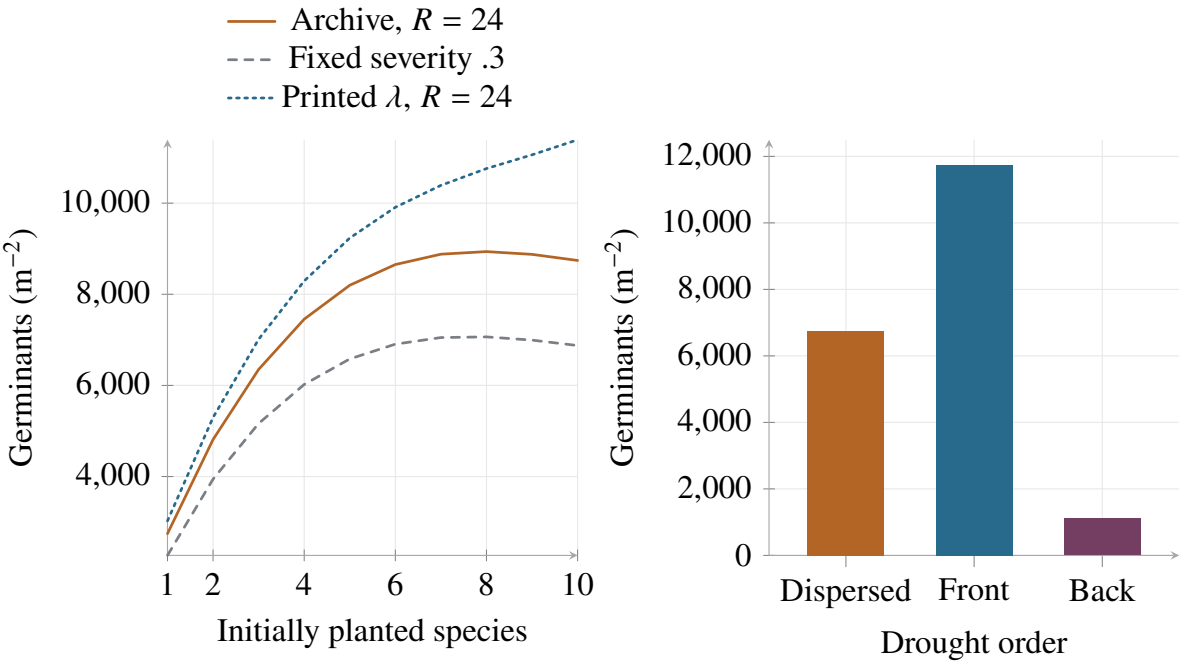


Figure 4. Parameter source and drought ordering

At constant interpolated severity .3, the ten-species stand averages 6,875 germinants/m², versus 8,415 under switching endpoints. Equal mean severity is not equal mean vital rates; this comparison tests the nonlinear scenario model and does not isolate variability experimentally. For the parameter-source comparison we regenerate an archive baseline with the same $R = 24$ weather matrix used by the printed-fecundity and interaction attacks. Its ten-species mean is 8,742; replacing only the drought fecundities raises it to 11,391 and makes the richness curve increase through ten species. The $R = 48$ principal curve is not used as this attack control. The archive supports saturation in the tested samples, but the precise high-richness maximum is source- and sample-sensitive. Scaling off-diagonal competition by ±20% under this same 24-weather design leaves SIAL as the sampled fifth-percentile candidate winner. These coefficient stresses are not confidence intervals because interaction standard errors are unavailable.

7 Pollution, habitat and a boundary to diversity insurance

7.1 A specified pollution mechanism

A generic pollution label does not identify a rate law. For a transparent toxic-fecundity stress, replace λ_i by $(1 - L)\lambda_i$, with $0 \le L \le 1$. The necessary rare-growth condition becomes

$$(1 - p)\log\{\beta_i(0) + (1 - L)\lambda_i(0)g_i(0)\} + p\log\{\beta_i(1) + (1 - L)\lambda_i(1)g_i(1)\} > 0. \quad (9)$$

The left side decreases strictly in L. A one-dimensional bisection finds each threshold, without fitting an unsupported dose-response curve. At $p = .3$, the allowable loss before isolated growth becomes nonpositive is 24.5% for BOOF, 47.5% for BRMA, 89.8% for VISA, 98.8% for MEPO and 99.1% for SIAL. These are mathematical fecundity-loss boundaries, not safe pollutant concentrations.

When every species has nonpositive rare growth, extra richness cannot restore persistence in this competitive model. Well below that boundary, live density can already deteriorate severely: at $L = .8$, the ten-species mean falls to 1,654; at $L = .95$, it is 357 germinants/m². Selection insurance does not replace pollution control. Nutrient pollution may instead favor fast-growing dominants and alter competition/asynchrony; field evidence for eutrophication weakening diversity-related stability motivates measuring those effects separately [6]. Our uniform toxicity scenario does not quantify nutrient enrichment.

7.2 Area loss and finite populations

Let a homogeneous habitat patch have area H and $M = H/a_0$ representative neighborhoods. Deterministic total abundance is $M\sum_i N_i$; reducing H alone leaves local density dynamics unchanged and reduces total abundance proportionally. Inserting area loss as an arbitrary extra per-capita mortality would conflate habitat amount with habitat quality.

For additional small-population effects we simulate integer seeds: germinants are binomial (n_i, g_i), dormant survivors binomial $(n_i - G_i, s_i)$, and new seeds Poisson with mean $\lambda_i G_i/[1 + \sum_j \alpha_{ij}G_j/M]$. We compare $.02, .2, 2, 20$ m² well-mixed patches with 256 trials, 120 years, $p = .6$, $\phi = .6$, and the same initial density. The denominator uses a self-inclusive mean-field closure: its $j = i$ term includes the focal germinant through the total G_i/M. The empirical coefficients describe surrounding neighbors; they do not validate focal-self inclusion or patch-edge effects at very small populations. Demographic independence, Poisson variance and this neighborhood closure are assumptions. These finite-patch risks are conditional model scenarios, not measured risks for real small habitats. The stochastic density drift need not equal the deterministic recurrence because germination counts enter a random denominator. An independent exact enumeration with $g = 1/2$, $s = 1/4$, $\lambda = 4$, $\alpha = 2$ and the same initial local seed density gives expected next densities $19/24$ for one seed in one neighborhood and $23/24$ for two seeds in two neighborhoods; plugging mean germination into the deterministic model gives $9/8$. This explains a finite-size drift effect in addition to extinction risk, rather than attributing every density difference solely to extinctions. Figure 5 contrasts these stress mechanisms. The left panel shows that uniform fecundity loss reduces live density despite increasing initial richness. The right panel shows extinction under joint 95% fecundity loss and clustered frequent drought,

with a strong patch-size effect. Neither axis denotes a measured pollutant dose or a validated fragmentation forecast.

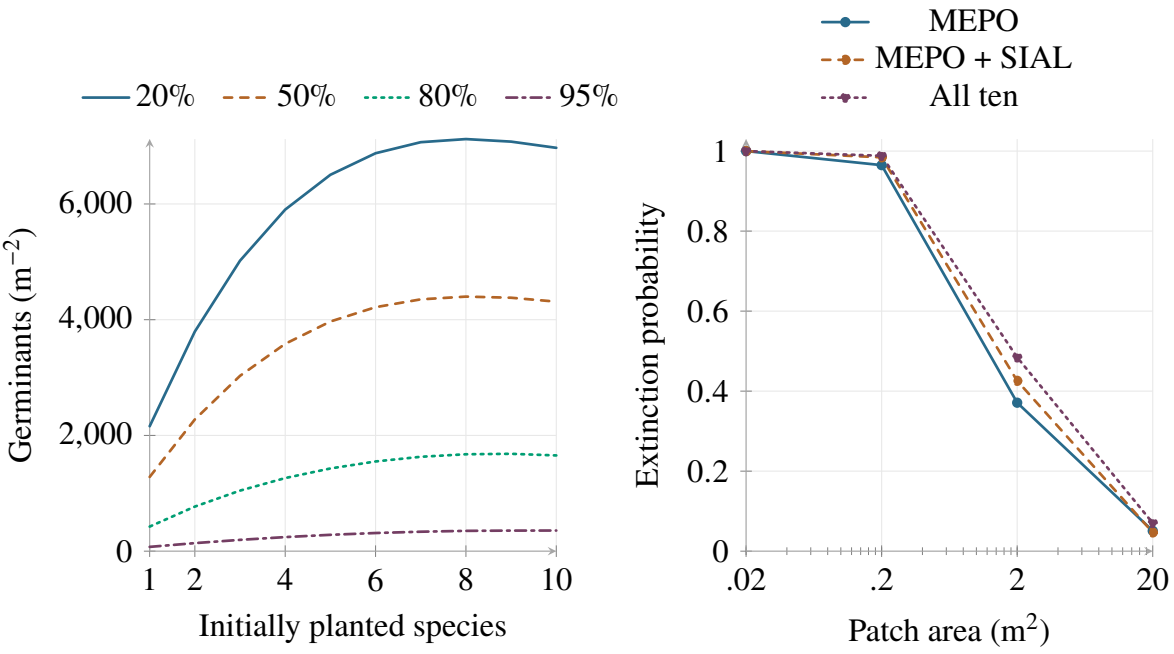


Figure 5. Pollution and habitat scenarios

No entire selected community became extinct in the 256 unpolluted trials at any tested area; zero observed events does not prove zero risk (the usual zero-event 95% upper bound is about $3/256 = 1.2\%$). Initially mixed communities generally ended with one species, and their full initial set was never retained. In the four-species set BRMA–MEPO–SIAL–VISA, mean density increased from 4,260 at the smallest area to 5,069 at 20 m². Competitive displacement, rather than complete stand extinction, dominates this parameterization. Connected refuges, immigration and real fragmentation geometry are outside this well-mixed area test.

A joint-stress attack changes that conclusion. With the same frequent clustered climate and a 95% fecundity loss, every tested community becomes extinct by year 120 in the .02 m² patches. At 2 m², extinction fractions are 37.1% for MEPO, 42.6% for MEPO–SIAL and 48.4% for all ten species; at 20 m², they are 5.1%, 4.7% and 7.0%. MEPO still has positive deterministic rare log growth under these rates, so that criterion is necessary but insufficient for finite-patch security. Loss of habitat and reproductive capacity combine to expose demographic failure that neither an unpolluted test nor a deterministic density equation establishes. The sampling standard error at a 48.4% estimate with 256 trials is 3.1 percentage points; this does not cover uncertainty in the ecological rate law.

8 Management for functioning and conserved communities

The central recommendation is to choose and protect response diversity, then verify coexistence. For uniformly sampled plantings from the ten-species pool studied here, four species meet the 90% goal for the illustrative 200-germinant/m² long-window floor under the moderate scenario; a 95% goal requires five. Before using either number in another pool, measure germi-

nation, viable seed production, seed survival and neighbor effects. A low-density growth screen excludes species that cannot replace themselves in the expected climate; reciprocal-invasion or resident-community invasion tests then screen competitive compatibility.

Our results support the following sequence of actions:

1. Select species with contrasting wet- and dry-season responses and a viable dormant stage. Do not equate high fecundity with a balanced mixture; SIAL and MEPO are density winners here but do not preserve the whole mixture.

2. Keep habitat mosaics and seed-bank refuges where different response types can reproduce. Spatial management is a proposed extension to the homogeneous model, not a simulated proof that a particular refuge layout works.

3. Reduce pollutant pressure and protect establishment-season moisture. Additional species cannot overcome a negative growth boundary shared by the entire pool.

4. Monitor annual germinants, reproductive output and species-specific viable seeds separately. Persistent total density with collapsing retained richness calls for composition intervention; a large seed bank with low live density calls for establishment intervention.

5. Re-estimate the rates after repeated drought, pollution or habitat change. Compare observed recruitment with the model and revise the mechanism when systematic errors appear.

Larger environmental benefits depend on maintaining living cover and ecological functions: soil protection, habitat provision and organic inputs may decline when germination collapses even if seeds remain. A community reduced to one dominant species can preserve some production while losing functional and food-web variety. We do not translate germinants into erosion rates, biomass carbon or pollinator service without local measurements. Managers should pair population monitoring with cover, bare-ground and soil indicators, rather than optimizing our density index alone.

9 Strengths, limits and conclusions

The model gives an explicit annual accounting, measured rate endpoints, an independent analytical baseline, exhaustive finite-pool comparisons and transparent stress mechanisms. The matched inputs prevent planting more total seeds from masquerading as a diversity benefit. The rare-growth boundary and area-scaling argument remain useful even if some empirical coefficients are revised.

The strongest limitations are the unresolved archive/article inconsistency, one-site endpoint calibration, absent field forecast validation, and omitted facilitation and spatial heterogeneity. The 48 weather trajectories quantify scenario variation, not parameter confidence or a 200-year prediction. The long-window floor smooths short catastrophic years; the added annual-minimum constraint shows that uninterrupted cover requires stricter composition insurance. The Poisson finite-patch extension needs calibration of demographic variation and dispersal.

We answer the problem with a conditional relationship: increasing initial richness improves composition insurance and often community density, with diminishing returns; it does not guarantee preservation of all species. Types and interactions determine whether a small targeted mixture outperforms random high richness. Drought frequency, clustering and chronological order change both population history and appropriate selection. Pollution can cross a growth boundary that diversity cannot repair; habitat loss lowers total abundance and can add demographic risk. Long-term management must preserve complementary viable populations and verify their coexistence, rather than treating an initial species count as a guarantee.

Reproduction protocol

The accompanying private bundle supplies `code/model.py`, `code/validate.py`, `raw/data/competition-csic.txt`, `run/output/results.json`, `run/output/traces.npz` and receipt/source snapshots. From the bundle root, run `python code/model.py --output replay/output`; the saved dependency receipt identifies the source environment (NumPy 2.5.3). No personal path is required by these instructions. The retained earlier run provides the validator's unchanged-main-result regression reference.

All random generators use NumPy `default_rng`, whose recorded bit generator is PCG64. For each deterministic simulation, independently restart seed 20231011 and draw one uniform array U of shape (T, R), in row-major order: $(200, 48)$ for principal scenarios, $(200, 24)$ for source/interaction/pollution attacks, and $(200, 1)$ for the fixed-severity case. Set $Z_{0,r} = 1[U_{0,r} < p]$ and then $Z_{t,r} = 1[U_{t,r} < p + \phi(Z_{t-1,r} - p)]$. A different R changes the sequence assigned to a replicate; seed equality alone is therefore not weather equality. Source and interaction attacks explicitly verify equality of their complete $(200, 24)$ weather arrays by SHA-256.

Enumerate masks $k = 1, \ldots, 1023$ using species position $i = 0, \ldots, 9$ in Table 2; species i is present when bit i of k is one. Each present species starts at $1000/S$ local seeds. For each year, record germinants from the pre-update seed bank, apply Equation (1) under that replicate's weather, and then record the updated bank and retained species. Average germinants across the final 50 years. Average equally over all subsets of a richness and weather replicates for the richness curve. A candidate's sampled fifth percentile is NumPy's default linear quantile across its R final-window averages ($R = 48$ for principal scenarios, $R = 24$ for source/interaction attacks); choose the first ascending-mask maximizer. This protocol reproduces the reported finite sample, not a population-optimal ranking.

For demographic tests use $T = 120$, $R = 256$ and weather seed 20231011 + 500. For each area and zero-based subset-list index k, restart demographic seed 20231011 + 300 + k. Each planted species starts with $\max(1, \text{round}(1000M/S))$ integer seeds, using Python's ties-to-even rounding; others start at zero. At each year draw the full $(256, 10)$ germination array, then dormant-survival array, then Poisson recruitment array, in that order; retain the self-inclusive denominator described in Section 7.2. The unpolluted subset list is VISA, MEPO–SIAL, BRMA–MEPO–SIAL–VISA, all ten; the joint-stress list is MEPO, MEPO–SIAL, all ten. Restart generators for each area. Extinction means zero total seeds after year 120; the final-30-year means are separate metrics. These restart and rounding conventions are part of the reproducible scenario, not empirical demographic rules.

References

[1] Matías, L., Godoy, O., Gómez-Aparicio, L. and Pérez-Ramos, I. M. (2018). An experimental extreme drought reduces the likelihood of species to coexist despite increasing intransitivity in competitive networks. *Journal of Ecology*, 106, 826–837. https://doi.org/10.1111/1365-2745.12962.

[2] Matías, L. et al. (2019). Data from the preceding study. Dryad, https://doi.org/10.5061/dryad.5d1s9. Actual downloaded CSIC mirror: https://digital.csic.es/handle/10261/183909. Competition-file MD5: `9f1026fe8c8a8f20d96cd17c6db653b6`.

[3] Pérez-Ramos, I. M. et al. (2019). Functional traits and phenotypic plasticity modulate species coexistence across contrasting climatic conditions. *Nature Communications*, 10, 2555. https://doi.org/10.1038/s41467-019-10453-0.

[4] Angert, A. L. et al. (2009). Functional tradeoffs determine species coexistence via the storage effect. *PNAS*, 106, 11641–11645. https://doi.org/10.1073/pnas.0904512106.

[5] Isbell, F. et al. (2015). Biodiversity increases the resistance of ecosystem productivity to climate extremes. *Nature*, 526, 574–577. https://doi.org/10.1038/nature15374.

[6] Hautier, Y. et al. (2014). Eutrophication weakens stabilizing effects of diversity in natural grasslands. *Nature*, 508, 521–525. https://doi.org/10.1038/nature13014.

[7] COMAP (2022). 2023 MCM/ICM Tips, V20221007, p. 5: minimum 12-point type. https://contest.comap.com/undergraduate/contests/mcm/contests/2023/problems/MCM-ICM_Tips.pdf.

[8] OpenAI Codex (2026-10-11). AI assistant used for research, model construction, implementation, mathematical checks, interpretation and manuscript generation. Host turn metadata reports `gpt-6.1-sol`; deployed service build and actual token/cost totals are unknown. See Report on Use of AI.

Report on Use of AI

This autonomous AI exercise used historical 2023 Problem A. No human independently solved or validated it; an AI subagent checked a specified implementation. The statement caps the entire report at 25 pages. Official 2023 Tips requires at least 12-point type [7]; normal text, captions and bibliography use 12 TeX points, without complete compliance certification. The summary year is 2023; demonstration number 7391856 is unassigned. This disclosure follows current 2027 guidance checked on 2026-10-11 at https://contest.comap.org/undergraduate/contests/mcm/instructions.php; historical AI rules remain unverified. The full PDF is below 25 pages.

Tool and purposes. OpenAI Codex (host identifier `gpt-6.1-sol`) read the authorized statement, compared models, retrieved primary ecology literature/data, wrote Python/LaTeX, interpreted results, implemented rational checks and inspected rendered pages. Python/NumPy supplied calculations, Matplotlib and PGFPlots figures, and Tectonic the build. Service version, tokens and cost are unavailable and not inferred from runtime.

Instruction and response record. Instructions required complete autonomous work toward Outstanding, all questions, learning boundaries, bounded computation, two seals and honest AI records. Equation (1), exhaustive comparisons and this report are the response. Before reading the question, the author's actual rollout was captured with the fixed export's `ai_records.py`: nine entry records, source hash and byte/line counts. Later same-thread snapshots preserve available prompts, responses and tool activity privately; they neither reconstruct missing history nor establish human verification.

Corrections and verification. Matrix-header parsing and validator JSON-interface failures remain recorded. Corrected baseline/autonomous runs passed 19/66 checks, including exact insurance and pollution roots. An independent AI review after sealing led to matched weather, reproduction details, sample qualifications and self-inclusive closure disclosure; feedback passed 68 checks. Both autonomous versions remain sealed. The archive/article conflict remains unresolved. The author read no same-problem solution, judge commentary or model answer during that stage. Pretraining is unknown; shared storage supplies process separation, not technical isolation.

Subsequent AI revisions organized shared assumptions, removed redundant heading levels, repaired section and figure navigation, and qualified summary rankings. These revisions retained model outputs and used separate whole-paper and final-size reviews; they add no human validation or new award assessment.

Attribution. Empirical inputs and scientific context are cited where used. Future weather, coefficient perturbations, management floors and habitat scenarios were created for this analysis and are not presented as observations. AI-generated interpretation has not been externally field-validated.